

SUPERMARKET SALES ANALYSIS

IBM SkillsBuild Data Analytics with AI Internship 2026

BharatCares in association with AICTE

Project Report

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B.Tech – Computer Science (Data Science)

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

1. Introduction

Supermarket Sales Analysis is a data analytics project focused on understanding sales performance, customer behavior, product performance, branch performance, payment methods, and customer ratings. The project uses Python-based data analysis and visualization to convert transaction data into useful business insights.

The analysis is based on 500 supermarket transactions containing information about invoice, date, branch, city, customer type, gender, product, category, quantity, unit price, payment method, customer rating, and sales amount.

2. Problem Statement

Supermarkets generate large amounts of transaction data. Without systematic analysis, it can be difficult to identify high-performing products, branches, categories, customer segments, and payment preferences. This project analyzes the available sales data to identify important trends and provide practical recommendations for business improvement.

3. Objectives

- Load and inspect the supermarket sales dataset.
- Check the dataset for missing or incorrect values.
- Calculate and verify sales using $\text{Quantity} \times \text{Unit Price}$.
- Analyze total sales, transaction count, quantity, and average transaction value.
- Compare sales performance across products and categories.
- Compare the performance of the four branches.
- Analyze payment method usage.
- Compare Member and Normal customer spending.
- Analyze customer ratings.
- Create visualizations to communicate the findings.
- Provide business recommendations based on the analysis.

4. Dataset Description

The dataset contains 500 supermarket sales transactions. Each row represents a transaction and contains the following fields:

Column	Description
Invoice ID	Unique transaction identifier
Date	Transaction date
Branch	Branch code
City	Branch city
Customer Type	Member or Normal
Gender	Customer gender
Product	Product purchased
Category	Product category
Quantity	Number of units purchased
Unit Price	Price per unit
Payment	Payment method
Rating	Customer rating
Sales	Transaction sales amount

5. Technologies and Libraries Used

Technology	Purpose
Python	Main programming language
Pandas	Data loading, cleaning, grouping and analysis
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Jupyter Notebook	Interactive development and presentation
GitHub	Project version control and submission

6. Methodology

1. Import the required Python libraries.
2. Load the supermarket sales CSV file using Pandas.
3. Inspect the dataset structure using head(), info(), shape and descriptive statistics.
4. Check for missing values and incorrect data types.
5. Convert numeric fields to appropriate numeric data types where required.
6. Verify sales values using Quantity × Unit Price.
7. Calculate totals, averages and counts.
8. Group the data by product, branch, category, payment method and customer type.
9. Create charts for comparison and interpretation.
10. Summarize the findings and prepare business recommendations.

7. Data Quality Check

The final project dataset contains 500 transaction records and 13 columns. The dataset was checked for missing values and numeric consistency before analysis. Sales values were also verified against Quantity × Unit Price.

Check	Result
Number of transactions	500
Number of columns	13
Missing values	None observed in the final dataset
Sales calculation	Verified using Quantity × Unit Price
Rating range	3.0 to 5.0 in the supplied data

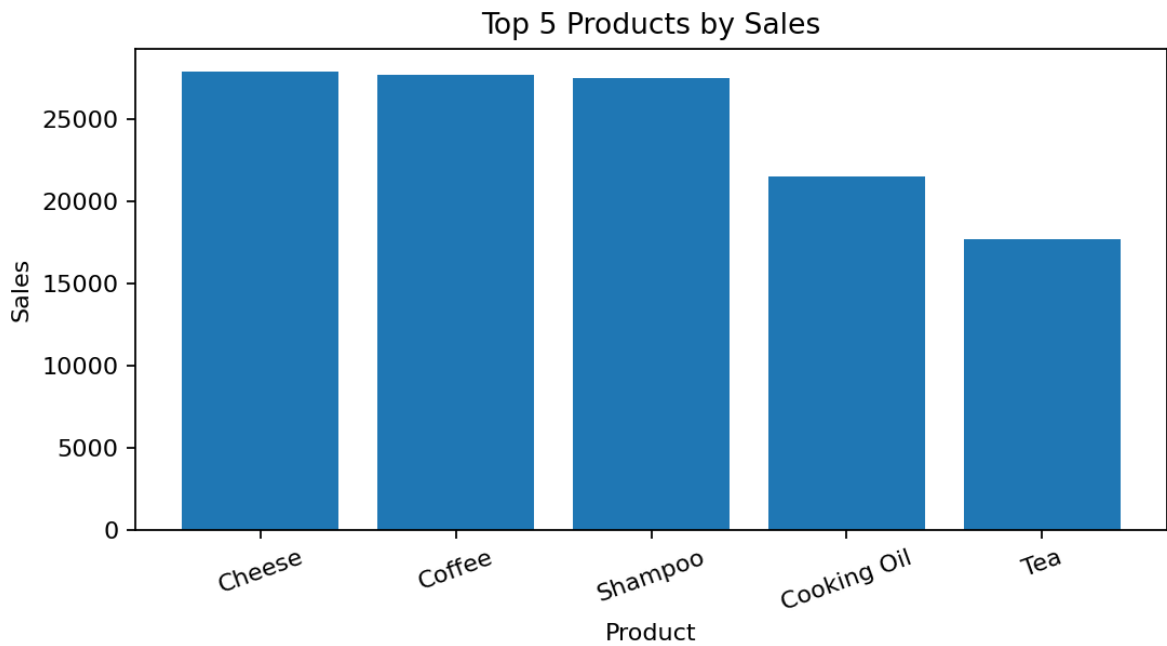
8. Overall Sales Summary

Metric	Result
Total transactions	500
Total quantity sold	2,737 units
Total sales	₹244,411.08
Average transaction value	₹488.82
Average customer rating	3.99 / 5

9. Product Analysis

Product-level analysis was performed by grouping transactions by product and summing sales. Cheese generated the highest sales among the products.

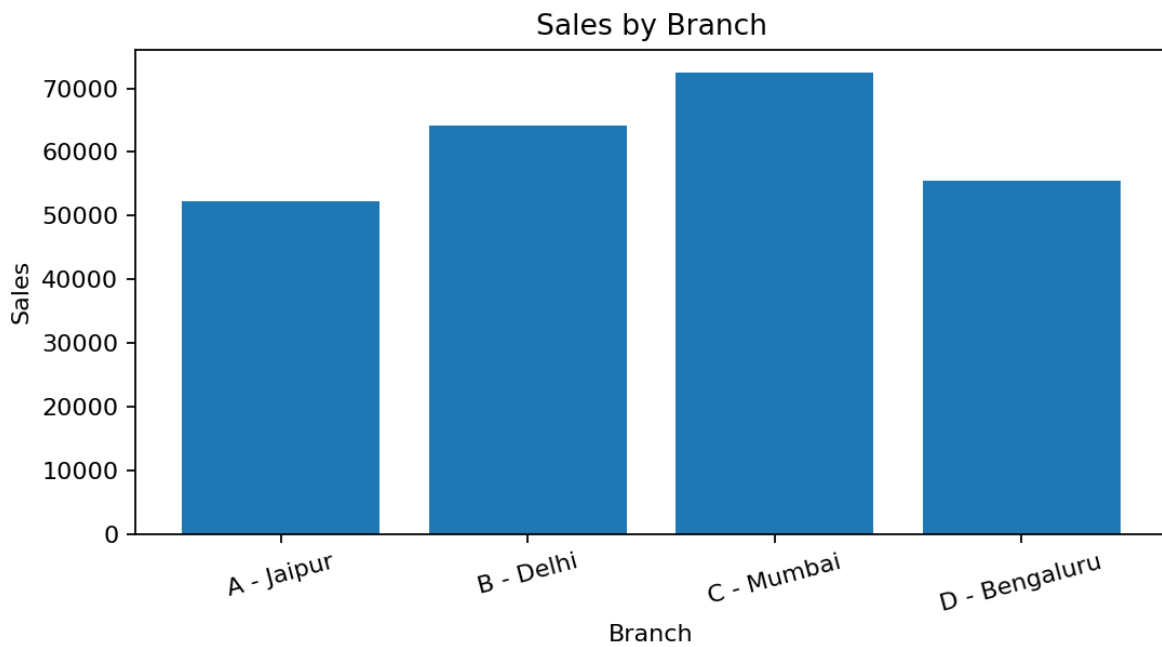
Rank	Product	Sales
1	Cheese	₹27,906.30
2	Coffee	₹27,694.87
3	Shampoo	₹27,497.48
4	Cooking Oil	₹21,525.06
5	Tea	₹17,681.59



10. Branch Analysis

Branch performance was compared using total sales. Branch C in Mumbai recorded the highest sales, making it the strongest-performing branch in the dataset.

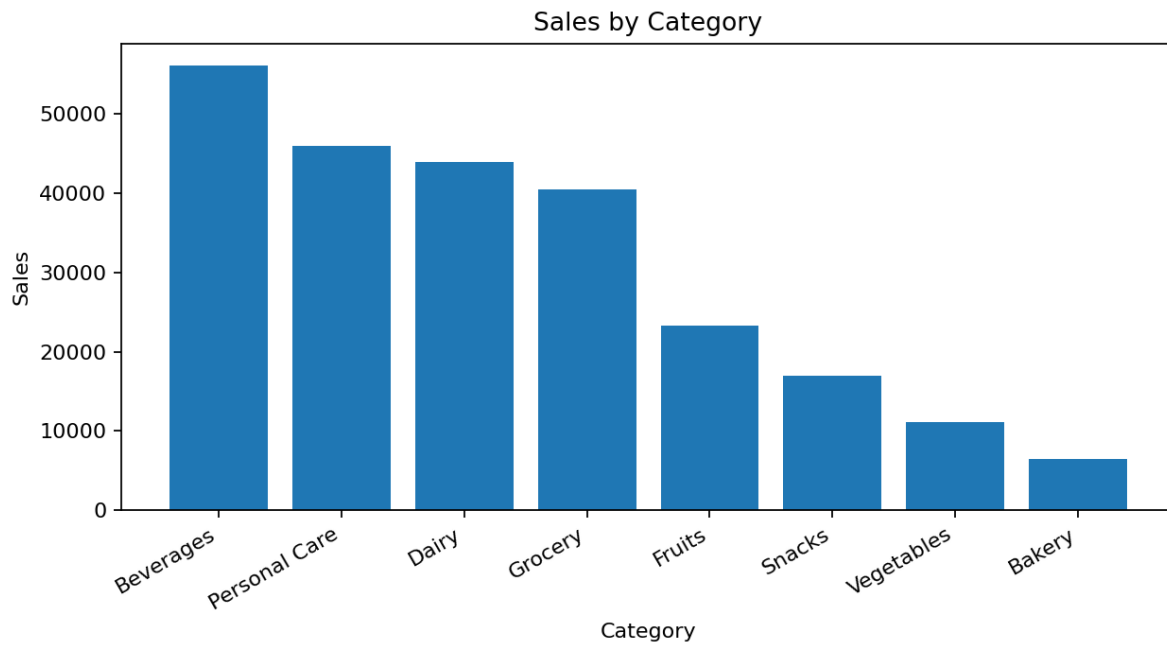
Branch	City	Sales
A	Jaipur	₹52,357.08
B	Delhi	₹64,116.26
C	Mumbai	₹72,469.45
D	Bengaluru	₹55,468.29



11. Category Analysis

Category-level analysis shows that Beverages is the highest-selling category. Bakery has the lowest sales among the listed categories.

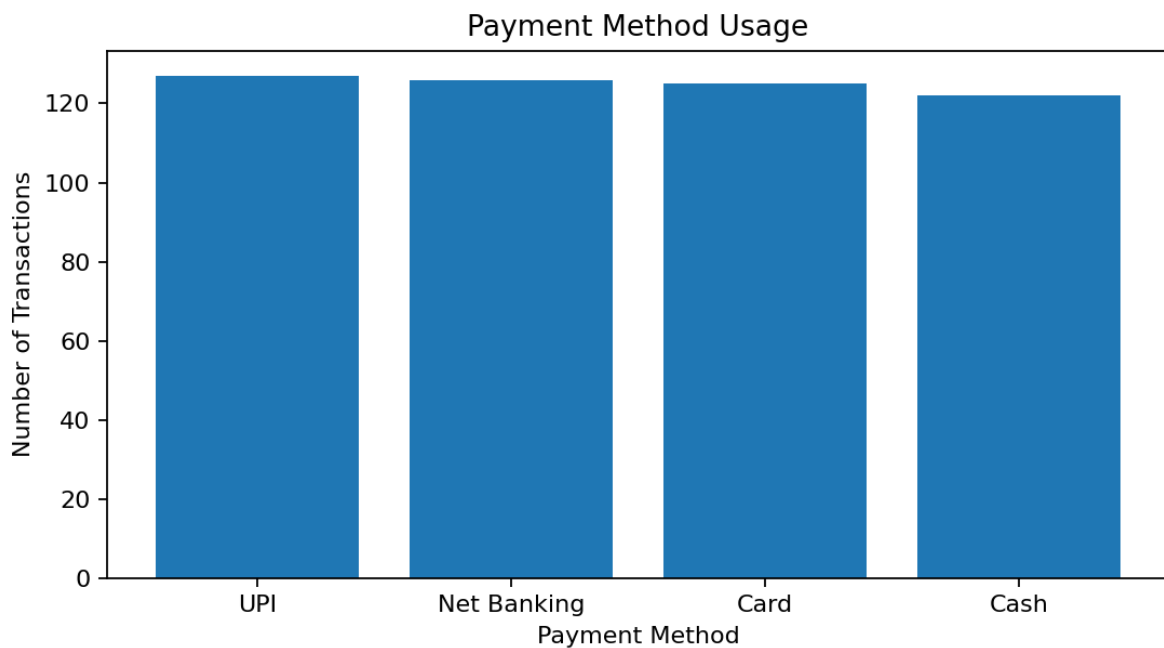
Category	Sales
Beverages	₹56,108.24
Personal Care	₹45,943.96
Dairy	₹43,992.00
Grocery	₹40,470.47
Fruits	₹23,263.17
Snacks	₹16,992.97
Vegetables	₹11,124.17
Bakery	₹6,516.10



12. Payment Method Analysis

UPI was the most-used payment method with 127 transactions. Net Banking and Card were also frequently used, while Cash had the lowest number of transactions among the four methods.

Payment Method	Transactions
UPI	127
Net Banking	126
Card	125
Cash	122



13. Customer Type Analysis

The average transaction value for Normal customers was slightly higher than for Member customers in this dataset. This indicates an opportunity to study the purchase behavior of Normal customers and identify ways to convert frequent Normal customers into members.

Customer Type	Average Transaction Value
Member	₹483.14
Normal	₹497.07

14. Customer Rating Analysis

The overall average customer rating is 3.99 out of 5. The ratings provide a useful indicator of customer satisfaction and can be used alongside product, branch and category performance to identify areas that may require service or quality improvements.

15. Key Findings

- Cheese is the highest-selling product with sales of ₹27,906.30.
- Branch C in Mumbai is the best-performing branch with sales of ₹72,469.45.
- Beverages is the highest-selling category with sales of ₹56,108.24.
- UPI is the most-used payment method with 127 transactions.
- Normal customers have a slightly higher average transaction value than Member customers.
- The average customer rating is approximately 3.99 out of 5.
- The analysis shows clear differences in performance across products, branches and categories.

16. Business Recommendations

- Focus on high-performing products such as Cheese, Coffee and Shampoo by maintaining availability and suitable promotions.
- Study the practices of Branch C in Mumbai and apply successful strategies to lower-performing branches.
- Promote the Beverages category because it generates the highest category sales.
- Continue supporting digital payments, especially UPI and Net Banking, because they account for a large share of transactions.
- Design targeted membership offers for Normal customers to encourage repeat purchases and membership conversion.
- Monitor customer ratings regularly and investigate products or services associated with lower ratings.
- Use product and branch-level sales trends for inventory planning and promotional decisions.

17. Conclusion

The Supermarket Sales Analysis project demonstrates how Python-based data analytics can be used to understand retail transaction data. The analysis identified the strongest products, best-performing branch, highest-selling category, payment preferences, customer segment differences and overall customer satisfaction.

The results can support practical decisions related to inventory, promotions, branch management, customer engagement and service improvement. The project also demonstrates the use of Pandas, NumPy, Matplotlib and Seaborn for a complete data analysis workflow.

18. Project Files

- VangaAkshaya_SupermarketSalesAnalysis.ipynb – Complete Jupyter Notebook
- supermarket_sales_data.csv – Dataset used for analysis
- requirements.txt – Python dependencies
- README.md – Project overview and setup information
- VangaAkshaya_SupermarketSalesAnalysis_ProjectReport.docx – This report

19. GitHub Repository

GitHub repository: <https://github.com/akshayavanga/Supermarket-Sales-Analysis>